

Întrebarea 543

Rezolvare

$$X_{30} = 23_{10}$$

$$u \equiv g^k \pmod{p}$$

$$u \equiv 3^3 \pmod{31}$$

$$u \equiv 27 \pmod{31}$$

$$v \equiv m \cdot \alpha^k \pmod{p}$$

$$\alpha^k \pmod{p}$$

$$\alpha^k \equiv 19^3 \pmod{31}$$

$$19^2 = 361 = 31 \times 11 + 20 \equiv 20 \pmod{31}$$

$$19^3 = 19^2 \cdot 19 \equiv 20 \cdot 19 = 380 \pmod{31}$$

$$380 = 31 \times 12 + 8 \equiv 8 \pmod{31}$$

$$v \equiv 23 \cdot 8 \pmod{31}$$

$$v \equiv 184 \pmod{31}$$

$$184 = 31 \times 5 + 29 \implies v \equiv 29 \pmod{31}$$

$$(u, v) = (27, 29)$$

$$27_{10} = ?_{30}$$

$$29_{10} = ?_{30}$$

Mesajul criptat este ?.